

Qatar-2b

R-1894

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_140413 · created 2026-08-03T13:15:23+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2456760.81646 +/- 9.4e-05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.196 +/- 0.022
Transit depth (Rp/Rs)^2	3.83 +/- 0.84 [%]
Orbital Inclination (inc)	87.12 +/- 1.55
Airmass coefficient 1 (a1)	1.102 +/- 0.033
Airmass coefficient 2 (a2)	-0.069 +/- 0.065
Scatter in the residuals of the lightcurve fit is	2.94 %
Best Comparison Star	None
Optimal Aperture	2.7
Optimal Annulus	10.18
Transit Duration (day)	0.0751 +/- 0.0043

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.196 $\pm$ 0.022 vs 0.16327 (deviation 1.36 σ)
Duration fit vs published	0.0751 d vs 0.0767 d (deviation 0.33 σ)
Mid-time O-C	0.4 min
Depth SNR	8.1
Residual scatter	2.94 %

Light curve

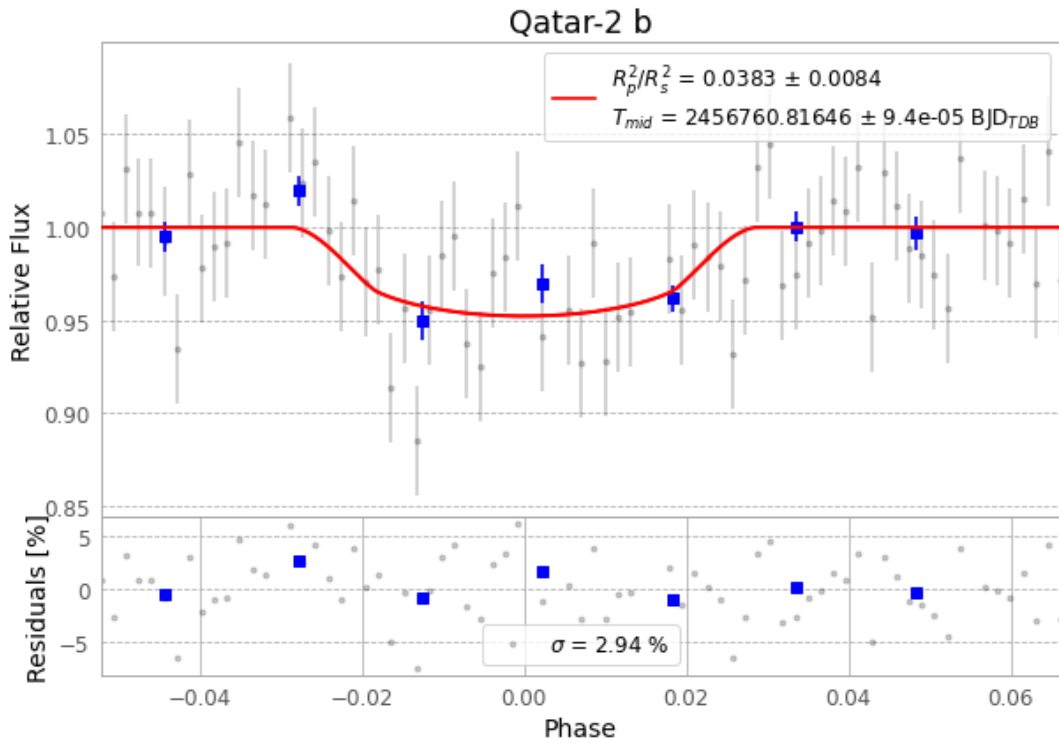


Figure 1 — FinalLightCurve_Qatar-2 b_2014-04-12.png (SHA-256 7dd2026447931262...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [205, 225], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 084c91d20bb2a7bd...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

75 FITS frames (49 MB), Qatar - 2140413054512.FITS → Qatar - 2140413093312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-140413.log (35137927024a...), FinalLightCurve_Qatar-2 b_2014-04-12.png (7dd202644793...), FinalLightCurve_Qatar-2 b_2014-04-12.csv (e4d376a2c595...)

Compute resources

Wall time 150.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 13:15Z.